

2

Internal Energy, Heat and Transfer Processes

In this chapter, we will learn about

- molecular kinetic energy and internal energy of a body
- energy transfer and thermal equilibrium
- conduction, convection and radiation as means of energy transfer
- microscopic interpretation of conduction
- factors affecting the emission and absorption of radiation

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2.1 Internal energy and energy transfer

Flipped classroom
Watch this and answer the questions.

DSE goal

Interpret temperature as quantity associated with average kinetic energy due to random motion of molecules in a system.

Relate internal energy to kinetic energy of random motion of molecules in the system.

Describe the effect of mass and temperature on internal energy of a system.

Simulation

→ This simulation compares the molecular arrangement and movement in different states of matter and temperatures. It is available on OUP web in PC version (.exe) only.

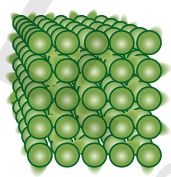
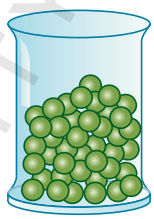
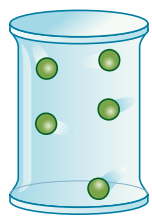
1 Internal energy

Recall the particle model of matter that we learned in junior science.

Pre-study **Particle model of matter**

According to the particle model of matter,

- matter is made up of tiny particles (atoms or molecules);
- the particles are constantly in random motion.

State of matter	Solid	Liquid	Gas
			
Molecular arrangement	Closely packed	Closely packed	Far apart from each other
Molecular movement	• Fixed in position • Vibrate about in fixed position	• Not fixed in position • Move around each other	• Not fixed in position • Move freely at high speeds in all directions

a Molecular kinetic energy

Caution

Note the difference between the KE of a body and the KE of the molecules in a body.

The molecules in a body possess **kinetic energy** (KE), so they are constantly in motion. The more kinetic energy the molecules possess, the faster they move.

The KE of a body is due to the movement of the body itself, and the KE of the molecules in a body is due to the random motion of the molecules.

A body at room temperature remaining at rest has zero KE, yet it has non-zero molecular KE as the molecules in it are moving.

The average kinetic energy of molecules in a body can be inferred by measuring the temperature of the body. When the molecules possess more kinetic energy and move more rapidly, the temperature of the body increases (Fig 2.1a on p.21).

kinetic energy 動能

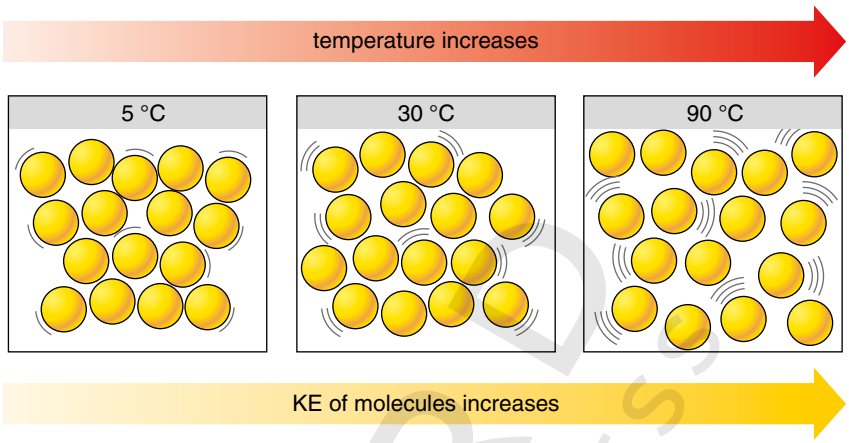


Fig 2.1a *How the KE of molecules changes with the temperature of a body.*

The above is supported by the following activity.

Video

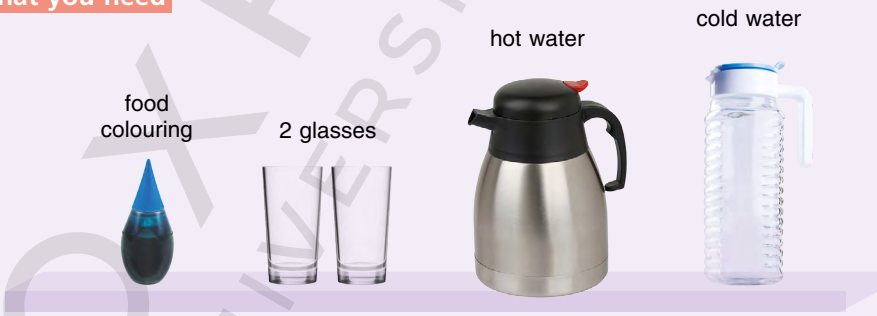
→ This video demonstrates the Hands-on activity.

Hands-on activity

→ Worksheet is available on OUP web.

Drops of food colouring in hot and cold water

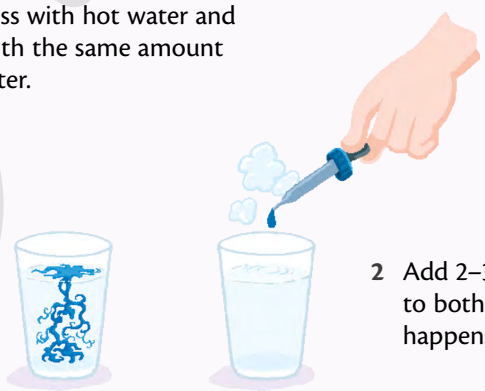
What you need



What to do

Beware of the hot water.

1 Fill one glass with hot water and another with the same amount of cold water.



2 Add 2–3 drops of food colouring to both glasses. Observe what happens to the food colouring.

What you find

In which glass of water does the food colouring spread more quickly?

In the **Hands-on activity**, the food colouring spreads faster in hot water (Fig 2.1b). It can be inferred that the molecules in hot water move more quickly to bring the molecules in the food colouring to different positions in a shorter time.

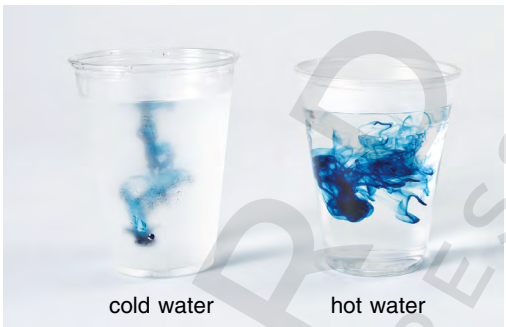


Fig 2.1b *Observation in the activity.*

The kinetic energy of a molecule is the energy that the molecule possesses due to its motion.

▶ The molecules in a body have a wide range of speeds and kinetic energies. In fact, the temperature is a measure of the **average** kinetic energy of molecules in the body. For two bodies with the same temperature, their molecules have the same average molecular kinetic energy.

✎ Ts may emphasize that temperature is NOT the average kinetic energy of the molecules. However, it is closely related to the average kinetic energy. It will be discussed in more detail in Ch 5.

The temperature of a body is a measure of the average kinetic energy due to the random motion of the molecules in a body.

✎ Ts may point out that the molecules of an object at 0 °C will still have 'plenty' of kinetic energy.

Note that the molecules in a body at 0 °C still have plenty of kinetic energy. The average kinetic energy of the molecules reaches the minimum value at about –273 °C. Nothing can be colder than this temperature.

Absolute zero will be further discussed in Ch 5.

▶ Therefore, this temperature is called **absolute zero**.

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Quick check 1

→ E-version in OUP Exercise Platform is also available.

1 *True or false:* If a body remains at rest, it has zero molecular kinetic energy. (T/F)

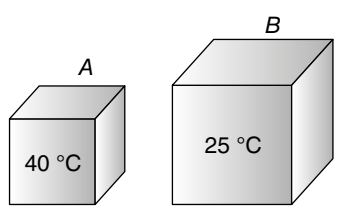
2 *True or false:* For a body at a certain temperature, all molecules in the body have the same kinetic energy. (T/F)

3 *True or false:* The temperature of a body is a measure of the total kinetic energy of its molecules. (T/F)

4 Consider iron blocks *A* and *B* as shown.

(a) *True or false:* The average kinetic energy of the molecules in *A* is higher than that in *B*. (T/F)

(b) *True or false:* Any molecule in *A* has a higher kinetic energy than any molecule in *B*. (T/F)



absolute zero 絕對零度

b Internal energy

Cross-link

The internal energy of a body also includes the total potential energy of all the molecules in the body. You will learn more about this in Ch 4.

The **internal energy** of a body is the total amount of the **molecular** energy stored in the body. The total kinetic energy due to random motion of molecules is a part of the internal energy. Like other forms of energy, the internal energy is measured in **joules** (J). Larger units of energy include the kilojoule (kJ) and the megajoule (MJ).

$1\ \text{kJ} = 10^3\ \text{J} = 1000\ \text{J}$

$1\ \text{MJ} = 10^6\ \text{J} = 1\ 000\ 000\ \text{J}$

Know more

Unit prefixes

For convenience, unit prefixes can be used to form units of various sizes. Some of them are listed below.

Prefix	m	c	k	M
Meaning	10^{-3}	10^{-2}	10^3	10^6

For example,

$1\ \text{km} = 10^3\ \text{m} = 1000\ \text{m}$

$1\ \text{kJ} = 10^3\ \text{J} = 1000\ \text{J}$

Note that these prefixes are case-sensitive. In particular, 1 mJ ≠ 1 MJ.

Since the internal energy of a body includes the total kinetic energy of all the molecules in the body, it increases with the temperature and the mass (which increases with the number of molecules) of the body (Fig 2.1c and 2.1d).

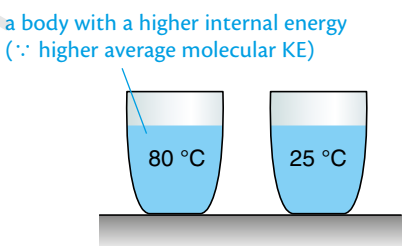


Fig 2.1c *Internal energy of a body increases with temperature.*

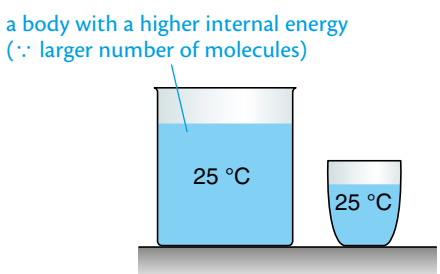


Fig 2.1d *Internal energy of a body increases with mass.*

internal energy 內能　　joule (J) 焦耳

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Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 Which of the following will increase if the temperature of a body increases?

- ☐ Average kinetic energy of the molecules in the body
- ☐ Total kinetic energy of the molecules in the body
- ☐ Internal energy of the body

(For Q2–3.) Two blocks *P* and *Q* are made of the same material and have the same mass (Fig a).

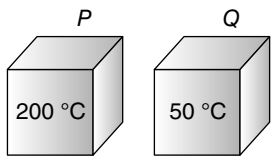


Fig a

2 In which block do the molecules have a higher average kinetic energy?

A *P*
B *Q*
C Same average kinetic energy
D Cannot be determined

3 Which block has a higher internal energy?

A *P*
B *Q*
C Same internal energy
D Cannot be determined

(For Q4–5.) Two blocks *P* and *Q* are made of the same material and *P* has a larger mass than *Q* (Fig b).

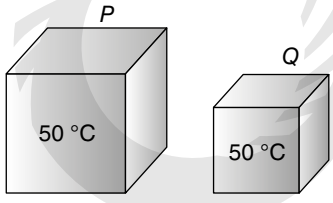


Fig b

4 In which block do the molecules have a higher average kinetic energy?

A *P*
B *Q*
C Same average kinetic energy
D Cannot be determined

5 Which block has a higher internal energy?

A *P*
B *Q*
C Same internal energy
D Cannot be determined

6 Two blocks *P* and *Q* are made of the same material and *P* has a smaller mass than *Q* (Fig c).

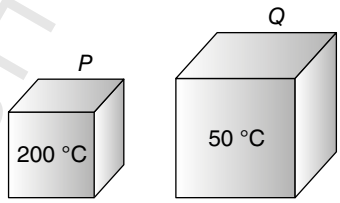


Fig c

Which of the following statements about the blocks must be correct?

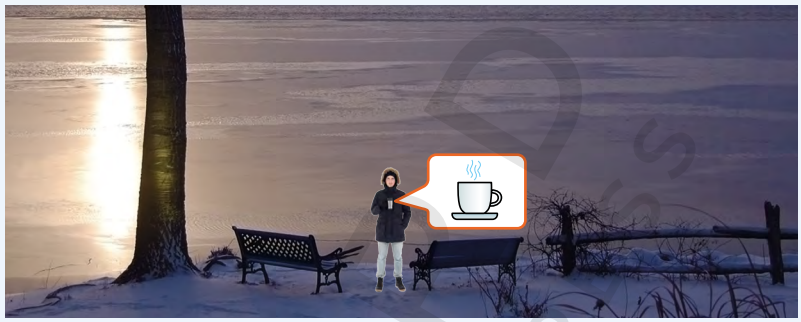
(1) The average kinetic energy of the molecules in *P* is higher than that in *Q*.
(2) The internal energy of *P* is higher than that in *Q*.

A None of the above
B (1) only
C (2) only
D (1) and (2)

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Right and wrong

Benny is enjoying a cup of hot chocolate in front of a frozen lake. Which of them, the frozen lake or the hot chocolate, is likely to have a higher total molecular KE?



✗ The hot chocolate has a higher temperature and therefore must have a higher molecular KE.

✓ The higher temperature of the hot chocolate only implies that it has a higher average molecular KE. As the frozen lake has a much larger mass, it has a much larger number of molecules. Therefore, the lake is more likely to have a higher total molecular KE.

DSE goal

Realize that heat is energy transferred due to temperature difference between two objects.

2 Energy transfer by heating

When some hot coffee is poured into a cup, energy is transferred from the hot coffee to the cup (Fig 2.1e). This process of energy transfer is due to the temperature difference between the two bodies and is called *heating*. The energy transferred in the process is called **heat**. We use the symbol Q to represent heat. Its unit is joule (J).

Many Ss confuse internal energy with heat. Internal energy is the energy stored in a body while heat is the energy transferred between two bodies as a result of a temperature difference.

Heat is the energy transferred from one body to another as a result of a temperature difference.

Note that the rate of energy transfer depends on the temperature difference. The larger the temperature difference, the higher the rate.



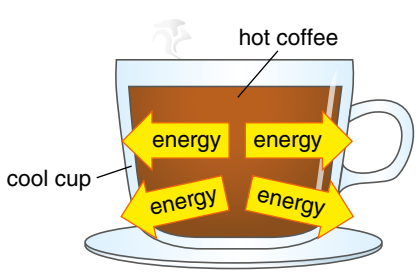


Fig 2.1e Transfer of energy from the coffee to the cup.

Know more

James Joule (1818–1889)

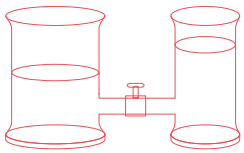
James Joule showed that 'heat' is a form of energy and not a substance.

▸ p.58

heating 加熱　heat 熱　James Joule 焦耳

✎ A clear distinction needs to be made among heat, internal energy and temperature.

Ts may use the following analogy to help Ss understand their differences.



The analogy is as follows:

container ⇒ object

amount of water ⇒ internal energy

water level ⇒ temperature

water flow ⇒ heat

Ts may ask Ss the following questions:

1 When will the water flow (heat flow) stop? When the amounts of water (internal energy) become equal or when the water levels (temperature) become equal?

2 Is it correct to say 'the container with higher water level (higher temperature) has more water (more internal energy)'?

✎ An activity to show the effect of doing work:

1 Half-fill a bottle with water and measure the temperature of the water.

2 Hold the bottle with a towel.

3 Shake the bottle for several minutes.

4 Measure the temperature of the water again.

An increase in temperature would be observed.

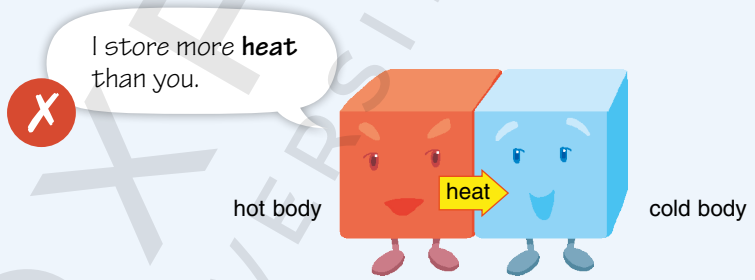
Heat flows naturally **from high-temperature bodies to low-temperature ones**. In Figure 2.1e on p.25, the coffee loses energy and its temperature decreases while the cup gains energy and its temperature increases. The energy transfer **stops when they reach the same temperature**. In this state, they are said to be in **thermal equilibrium**.

Right and wrong



✗ 'Coldness' is transferred from a cold body to a hot body.

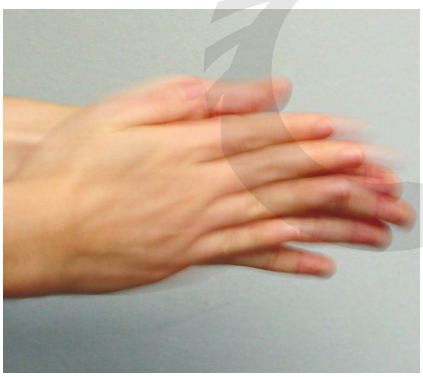
✓ In physics, only heat (energy) is transferred from a hot body to a cold body. There is no such a quantity called 'coldness'.



✗ Heat describes the energy stored in a body.

✓ Heat refers to the energy that **transfers** between two bodies. It is **not** the energy stored in a body. We use internal energy to describe the energy stored in a body.

Besides heating, doing **work** is another way to increase the internal energy of a body. The following are some examples of doing work (Fig 2.1f).



(i) Rubbing hands together.



(ii) Shaking a bottle of water.



(iii) Hitting a metal with a hammer.

Fig 2.1f *Examples of doing work.*

thermal equilibrium 熱平衡　　work 功

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Quick check 3

→ E-version in OUP Exercise Platform is also available.

1 *True or false:* Temperature is transferred from a hotter body to a cooler body. (T/F)

2 *True or false:* Heat flows from an object of higher temperature to an object of lower temperature. (T/F)

3 *True or false:* Heat flows from an object with higher internal energy to an object with lower internal energy. (T/F)

4 *True or false:* The energy transferred to an object due to a temperature difference is stored as heat. (T/F)

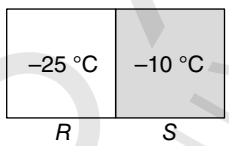
5 A metal rod is at room temperature. Which of the following actions can increase the internal energy of the metal rod? Put a tick in the corresponding box(es).

- ☐ Heat the metal rod with fire
- ☐ Put the metal rod in ice water
- ☐ Rub the metal rod with towel
- ☐ Hammer the metal rod

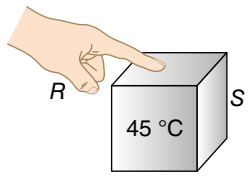
Learn by practice 1

1 In each of the following cases, draw an arrow to show the direction of energy transfer between *R* and *S*.

(a)



(b)



2 State how heat flows when a boy touches a piece of hot pizza.

3 State how heat flows when a girl touches a piece of ice.

Exercise 2.1

★**1**



Fig a

When we put a cold spoon into a bowl of hot soup (Fig a),

A all the internal energy of the soup is transferred to the spoon.

B the internal energy of the soup decreases.

C the spoon stores more heat than before.

D the temperature of the spoon decreases.

★**2** Which of the following statements is/are correct?

(1) The average kinetic energy of the molecules in a body reaches the minimum value when the body is at 0 °C.

(2) If two bodies are at the same temperature, the molecules in the two bodies will have the same average kinetic energy.

(3) If the molecules in two bodies have equal total kinetic energy, their average kinetic energy will also be equal.

A (1) only

B (2) only

C (1) and (2) only

D (2) and (3) only

★**3** Consider a jar of cold water and a glass of hot water. Which of the following statements must be correct?

(1) If they are put in contact, energy will be transferred from the hot water to the cold water.

(2) The kinetic energy of any molecule in the hot water is higher than that in the cold water.

(3) The average kinetic energy of the molecules in the hot water is higher than that in the cold water.

A (3) only

B (1) and (2) only

C (1) and (3) only

D (1), (2) and (3)

★**4** Which of the following ways can always increase the internal energy of an object?

(1) Put the object in contact with another one with a higher internal energy.

(2) Heat the object.

(3) Rub the object vigorously.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

★**5** Glasses *P* and *Q* contain the same amount of oil while glass *R* contains more oil (Fig b).

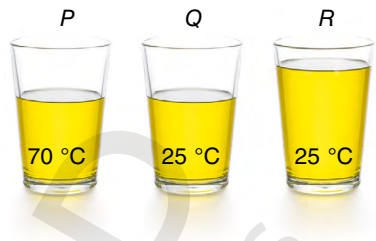


Fig b

(a) Which glass of oil has the highest average molecular kinetic energy?

(b) State and explain the direction of heat flow between the glasses in each of the following cases.

(i) Putting glasses *P* and *Q* in contact

(ii) Putting glasses *Q* and *R* in contact

★**6** Mary feels cold. She warms her hands by holding a cup of hot drink (Fig c).



Fig c

(a) State the direction of heat flow between the hand and the cup.

(b) After a short time, the hot drink cools down. Describe the change in the average molecular kinetic energy and the internal energy of the hot drink.

(c) Mary can also warm her hands by rubbing them. Why does it work?

★**7** A metal block at 80 °C is put into a pot of water at 20 °C.

(a) State and explain the change in temperature of the water.

(b) State the change in the internal energy of the water.

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2.2 Heat transfer processes

DSE goal
Interpret energy transfer by conduction in terms of molecular motion.

1 Conduction

If a metal spoon is put inside a bowl of hot soup (Fig 2.2a), the tip in direct contact with the soup will become hot in a short time. The handle of the spoon will also be gradually warmed up. Both situations result from heat transfer by **conduction**.

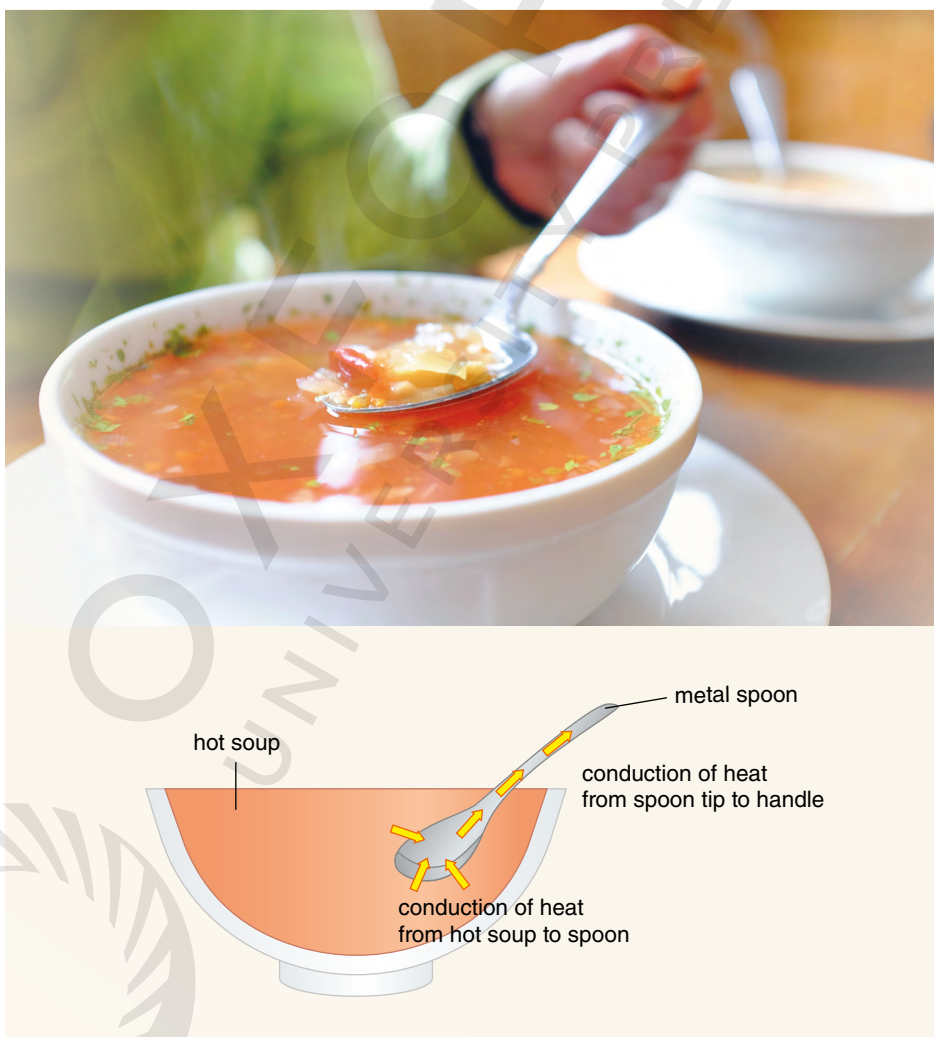


Fig 2.2a *Heat transfer by conduction.*

Conduction is the transfer of heat from the hot part to the cold part of an object, or from a hot object to a cold object in physical contact. Energy (but not matter) is transferred in the process.

conduction 傳導

Mercury is a liquid (at room temperature) that conducts heat well. It can respond to temperature changes quickly. This makes it a suitable liquid to be used in liquid-in-glass thermometers.

a Heat conductivity

→ This is a Video Quiz about the experiment 'Conduction in solids, liquids and gases'.

Different materials have a large difference in their abilities to conduct heat (Fig 2.2b). For example, silver conducts heat over 10 000 times faster than air does.



Fig 2.2b *Comparison of the ability to conduct heat between different materials (relative to glass).*

We can see that:

1. Metals are generally better **conductors** of heat than non-metals.
2. Non-metallic solid, liquids and gases are generally poor conductors, or good **insulators**.

However, there are some exceptions. For example, graphite, which is a special form of carbon and a non-metal, can conduct heat faster than many metals.

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Quick check 4

→ E-version in OUP Exercise Platform is also available.

1 *True or false:* Conduction can take place between objects in direct contact. (T/F)

2 *True or false:* Conduction can take place within an object. (T/F)

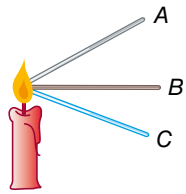
3 During the process of conduction, how is energy transferred?

A From a higher position to a lower position

B From a part with a higher internal energy to a part with a lower internal energy

C From a part with a higher temperature to a part with a lower temperature

4 Rods *A*, *B* and *C* are made of iron, copper and glass respectively. Which rod will heat up first if the three rods are of the same size?



5 Determine whether each of the following substances is a poor conductor of heat. Put a tick in the corresponding box(es).

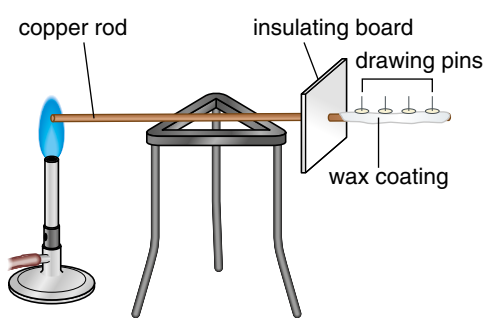
☐ Copper ☐ Plastic ☐ Water ☐ Air

Learn by practice 2

1 In the figure, one end of the copper rod is heated.

(a) Draw an arrow in the figure to show the direction of heat flow by conduction within the rod.

(b) The wax coated on the rod will melt when it becomes hot. Will the drawing pins drop at the same time? Explain your answer.



→ This video shows this demonstration. Video

conductor 導體　　insulator 絕緣體

b Examples of conduction and insulation

Both conductors and insulators are important to us and are commonly found in daily life. A conductor allows heat to flow through it quickly so that it can be used in heating up or cooling down an object. On the other hand, heat flows slowly through an insulator, which can therefore be used to reduce the temperature change of an object.

Writing practice

Question

One way to thaw (defrost) a piece of frozen meat is to put it on a metal pan (Fig a). Explain why this method helps to speed up the thawing process. (2 marks)

Tip

You may write the explanation by answering the following questions:

① Which material is responsible for the conduction? Is it a good or poor conductor?

② How does it affect the rate of conduction?

③ What is the direction of heat transfer?

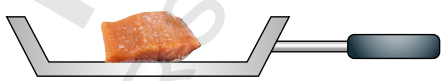


Fig a

Answer

Metal is a good conductor of heat. 1A

Therefore, heat is conducted quickly from the metal pan to the frozen meat. 1A

Practise yourself

1 Some ice is kept under water at the bottom of a boiling tube with a piece of wire gauze (Fig b). The top part of the water is then heated. When this part of water starts boiling, most of the ice remains unmelted. Explain the reason. (2 marks)

If the ice is allowed to float on the top, it will melt before the water boils. This is because heat is transferred from the lower end to the upper end through convection.

Video

→ This video shows this demonstration.

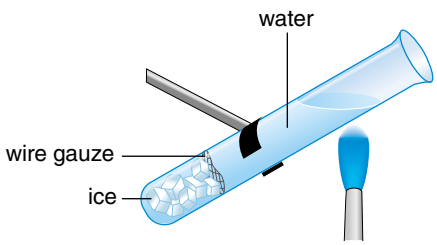


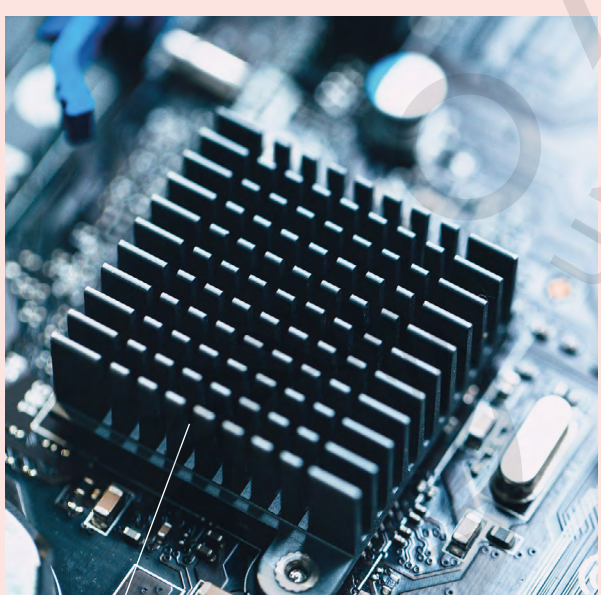
Fig b

Figure 2.2c shows some examples of enhancing and reducing conduction.

Enhancing conduction



frying pan
It enhances the heat gain by the food. It is made of metals which are good conductors so that heat is conducted to the food quickly.



heat sink
It enhances the heat loss of the processor. It is made of metals which are good conductors so that heat is conducted to air quickly to cool down the processor behind.

Reducing conduction



oven glove
It reduces the heat gain by the hand wearing it. It consists of heat-insulating materials which are poor conductors so that heat is conducted to the hand slowly.



polar bear's hair
It reduces the heat loss of the bear's body. Air is tapped between and inside the hair. Air is a poor conductor so that heat is conducted away from the body slowly to keep the bear warm.

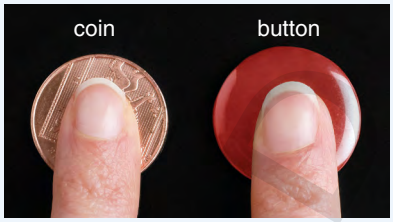
See p.33 for more information.

Fig 2.2c *Examples of enhancing and reducing conduction.*

A polar bear has a thick layer (about 10 cm) of insulating fat beneath its skin. Moreover, its fur makes the best use of the insulating property of air. The fur is made up of two types of hairs. The fine insulating hair near the skin acts rather like underwear and traps warm air close to the skin. The oily and transparent hollow long hair (as described in Fig 2.2c on p.32) can also guide sunlight to be collected and absorbed by the black skin of the bear.

Right and wrong

Linda touches a metal coin and a plastic button at the same time. She feels that the coin is colder.



✗ The coin must have a lower temperature than the button.

✓ An object feels cold because our energy is transferred to it. Metal is a better conductor than plastic. Energy is conducted away from Linda's finger to the metal coin more quickly than to the plastic button even if the coin and the button have similar temperatures.

Video → This video shows more interesting demonstrations about this phenomenon.

Learn by practice 3

1 Ken carefully moves a wooden splint until it is about 0.5 cm away from a Bunsen flame (Fig a). He finds that although the flame has a very high temperature, the wooden splint does not light up. Explain the result.

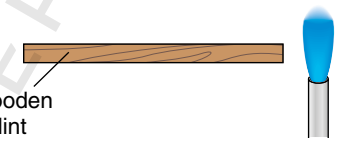


Fig a

2 Figure b shows a wok.



Fig b

(a) How does the metal body enhance cooking?

(b) Explain why the handle is made of wood instead of metal.

3 The same amount of ice cubes are placed on a metal plate and a wooden plate (Fig c). On which plate do the ice cubes melt more quickly? Explain your answer.

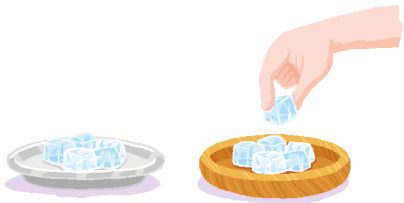


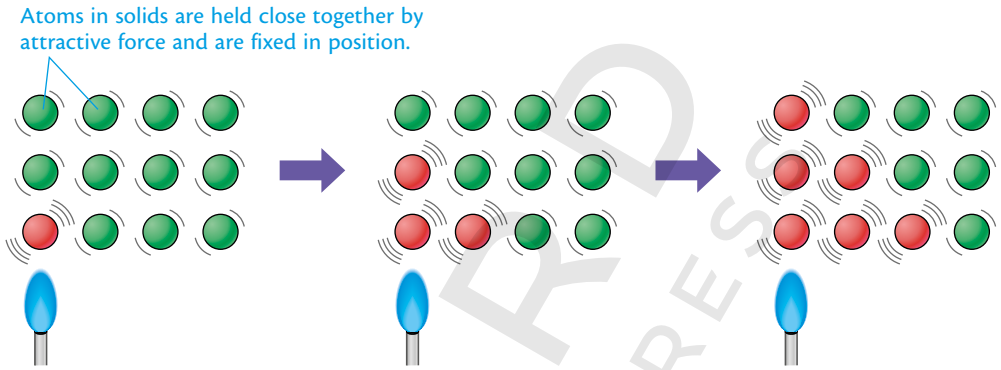
Fig c

C Conduction and particle motion

Simulation

→ This simulation shows the particle motion in a metal bar as heat is conducted from the hot end to the cold end.

The conduction of heat can be explained by the particle model. Consider a solid being heated at one end (Fig 2.2d):



(i) Atoms in solids are free to vibrate. When one part of a solid is heated, the atoms there vibrate faster.

(ii) The fast vibrating atoms hit the slower neighbouring atoms, making them vibrate more rapidly.

(iii) The process continues to spread out. Energy is passed, from atom to atom, to other parts of the solid.

Fig 2.2d *How heat is conducted in a solid.*

Metals conduct heat much faster than most non-metallic solids. To explain this, we need to first review some basic concepts about the atomic structure.

In junior science, we have learned that in each atom, there are some electrons moving around a nucleus (Fig 2.2e).

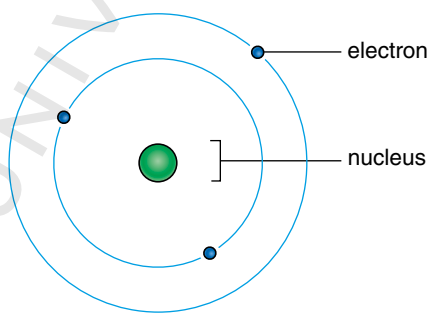


Fig 2.2e *Atomic structure.*

Different from atoms of most non-metallic solids, the outermost electrons of the metal atoms are loosely held. These electrons can move freely among the atoms (Fig 2.2f). They are called **free electrons**.

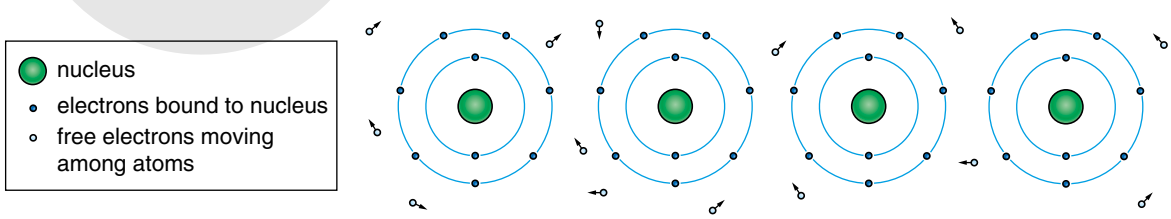


Fig 2.2f *Free electrons in metal.*

free electron 自由電子

Free electrons are also involved in the heat conduction of metals. When one part of a metal is heated, the free electrons there speed up and collide with other atoms and electrons (Fig 2.2g). Therefore, energy can be passed to other parts very quickly.

The rate of conduction depends on the density of free electrons. A good conductor has plenty of free electrons per unit volume. A poor conductor has almost none.

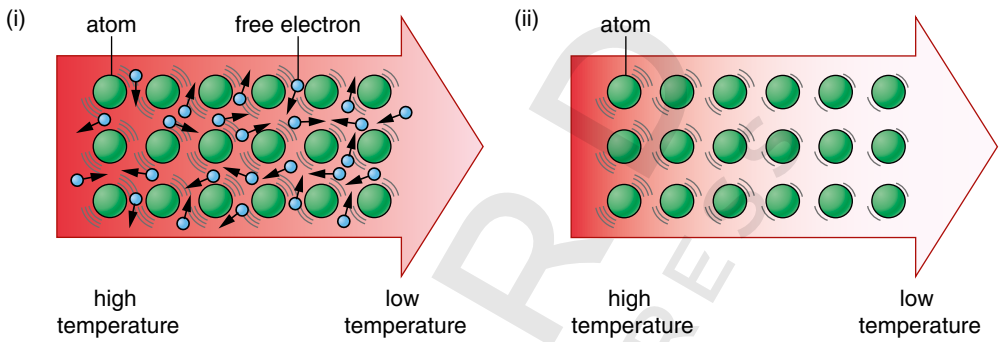


Fig 2.2g *Conduction (i) in a metal and (ii) in a non-metal.*

Simulation

→ This simulation demonstrates microscopic description of conduction in different materials. It is available on OUP web in PC version (.exe) only.

Google Forms

Microsoft Forms

Non-metallic liquids and gases are, in general, poor conductors.

Molecules in gases are much further apart than those in solids and liquids. Faster molecules in gases are less likely to collide with other molecules. As a result, gases are the poorest conductors (i.e. the best insulators).

Quick check 5

→ E-version in OUP Exercise Platform is also available.

1 *True or false:* Most metals are good conductors because they have many electrons moving around a nucleus. (T/F)

2 *True or false:* Metal only conducts heat through collisions of free electrons with neighbouring electrons. (T/F)

3 *True or false:* Gases are poor conductors because gas molecules are far apart. (T/F)

4 When one end of a metal bar is heated, which of the following happens? Put a tick in the corresponding box(es).

- ☐ Energy is transferred from the heated end to the other end.
- ☐ Atoms at the heated end vibrate more vigorously.
- ☐ The fast-moving atoms at the heated end move to the other end.

Learn by practice 4

1 Explain how a solid conducts heat by considering the particle motion in it.

2 Explain why a metal conducts heat quickly by considering the particle motion in it.

3 Explain why a gas conducts heat slowly by considering the particle motion in it.

2 Convection

As mentioned in earlier section, water (non-metallic liquid) is a poor conductor of heat. However, when the bottom of a kettle of water is heated, it only takes a short time for the water to start to boil. In this case, heat is transferred in water mainly through another process called **convection**.

a Convection in liquids and gases

As we have learned in junior science:

Convection only occurs in fluids since fluids do not have fixed shape and can flow from one position to another.

1 Convection occurs in fluids, i.e. liquids (Fig 2.2h) or gases (Fig 2.2i), but not in solids. Molecules of fluids can move around easily.

2 During the process, a hot fluid rises. **Heat is transferred upwards by the movement of the fluid itself**. The cold fluid in the surrounding region then moves in to replace the hot one.

Simulation

→ This simulation demonstrates the motion of convection currents in water. It is available on OUP web in PC version (.exe) only.

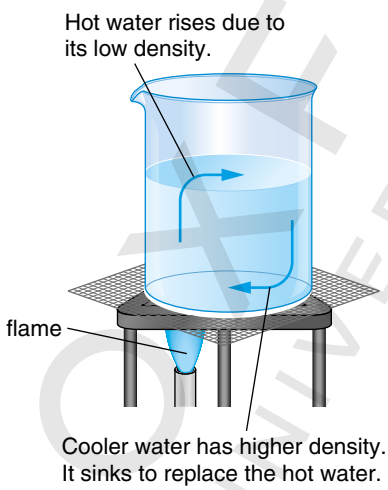


Fig 2.2h Convection in water.

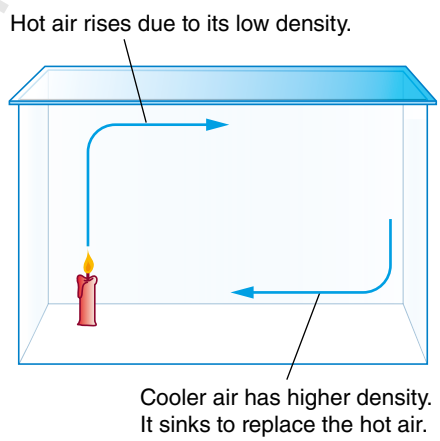


Fig 2.2i Convection in air.

Convection involves a transfer of matter but conduction does not.

Convection is the process of heat transfer in a fluid (a liquid or a gas) by the actual movement of the fluid itself.

The flow of fluid due to convection is called the **convection current**.

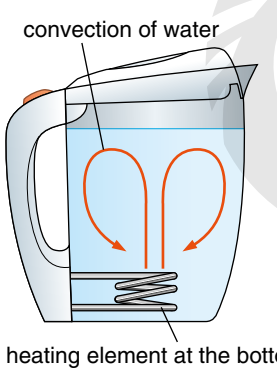


Fig 2.2j The design of enhancing heat flow by convection in an electric kettle.

b Examples of convection

Many appliances, such as electric kettle (Fig 2.2j) and air conditioner, are designed to enhance heat flow by convection.

Convection also occurs naturally. Sea breezes and land breezes, which are commonly observed in coastal areas in summer, are caused by convection (Fig 2.2k & 2.2l on p.37).

convection 對流　　convection current 對流

Cross-link

The reason why the temperature of land changes faster than that of the sea will be explained in Ch 3.

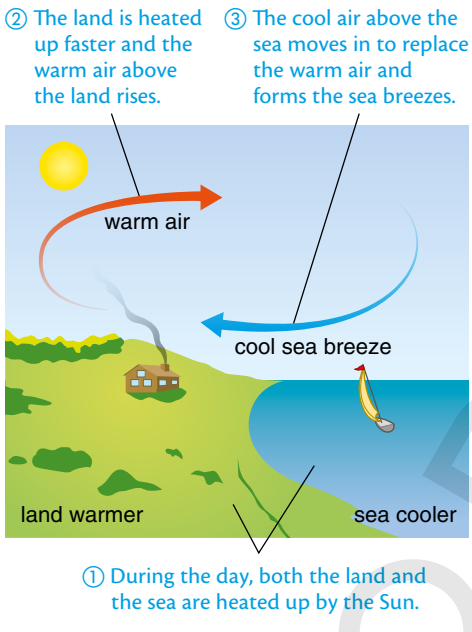


Fig 2.2k *How a sea breeze is produced during the day.*

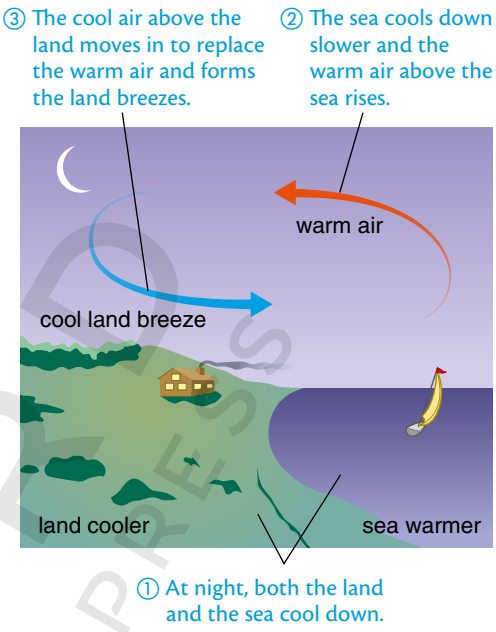
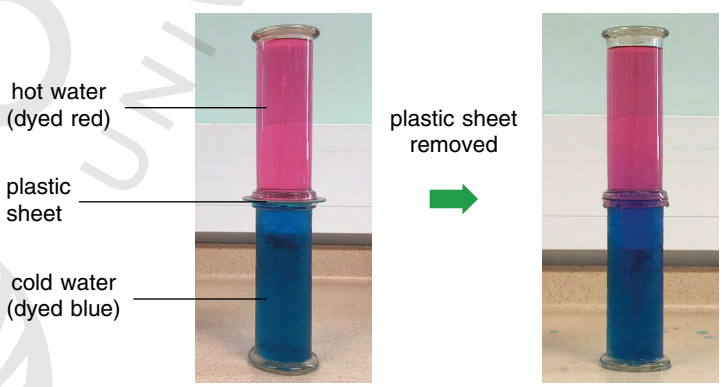


Fig 2.2l *How a land breeze is produced during the night.*

Example 1 Can the water mix?

Ivy places two glass columns of hot and cold water as shown. She thinks that when the plastic sheet between the two columns is removed, hot and cold water will mix immediately due to convection. However, actually the water does not mix well completely until the water reaches thermal equilibrium after a period of time. Explain why.



Solution

Since the hot water is already at the top, it cannot rise even when the plastic sheet is removed. Similarly, the cold water at the bottom cannot sink. As a result, no convection current can be formed.

When demonstrating this activity, Ts are suggested to prepare two set-ups at the same time for comparison:

Set-up 1: Hot water placed on top of cold water

Set-up 2: Cold water placed on top of hot water

The temperature of hot water needs not to be very high. Water of 60 °C and 0 °C has a temperature difference that is large enough to show the result.

Think about...

What will happen if an air conditioner is installed at a low position in a room?

Heat can be transferred quickly by convection in air. If we need to slow down this heat flow, we may need to limit the space in which air flows. The following are some examples (Fig 2.2m).



Fig 2.2m *Examples of reducing heat flow by convection.*

Google Forms Microsoft Forms

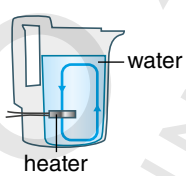
Quick check 6

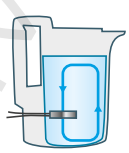
→ E-version in OUP Exercise Platform is also available.

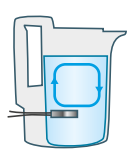
1 Can convection occur in the following media? Put a tick in the corresponding box(es).

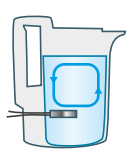
☐ Carbon dioxide gas ☐ A vacuum ☐ Oxygen gas
☐ A copper block ☐ Tea ☐ A book

2 An immersion heater is fixed in the middle level of a kettle. Which of the following diagrams correctly shows the convection current?

A 

B 

C 

D 

Learn by practice 5

1 When buying a box of ice cream mooncakes, a pack of dry ice (at about –79 °C) is usually placed on top of the box to keep the mooncakes cold (Fig a). Suggest a reason why the dry ice is placed on top of the box but not below it.

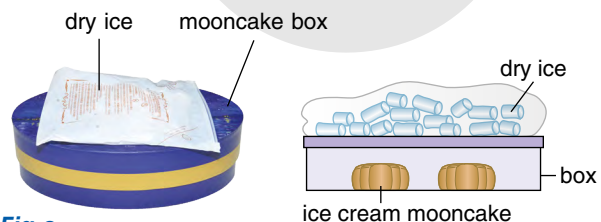


Fig a

2 In Chinese herbal tea shops, bowls of herbal tea are usually covered by glass plates (Fig b). Explain why this helps to keep the tea warm.



Fig b

3 Radiation

We can feel the energy transferred from the Sun to us (Fig 2.2n). However, the space between the Sun and the Earth is mostly empty. Conduction and convection cannot take place in a vacuum. So, how is the energy transferred in this case?



Fig 2.2n *Feeling the energy from the Sun.*

The energy transfer from the Sun to the Earth is done by **radiation**. The following are some basic concepts of radiation that we have studied before.

1 Radiation is a process of heat transfer which can take place in a vacuum (Fig 2.2o). It does not require any medium or particles to transfer energy.

2 Radiation is emitted in all directions (Fig 2.2p).

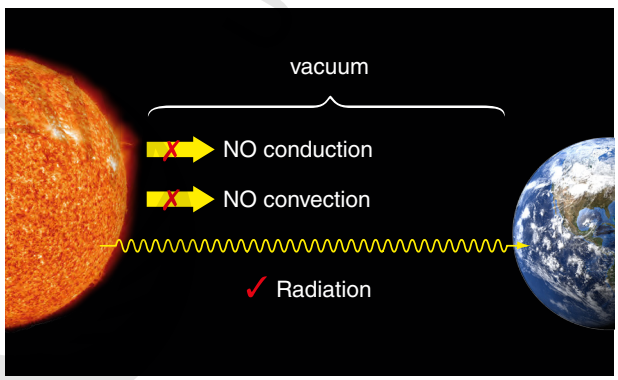


Fig 2.2o *Energy from the Sun has to travel through a vacuum before reaching the Earth.*

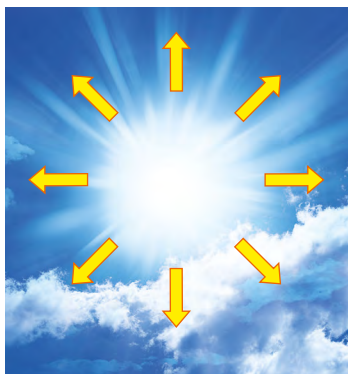


Fig 2.2p *The Sun emits radiation in all directions.*

Radiation is a process of heat transfer which does not require any medium or particles. It takes place in all directions.

Know more

Meanings of radiation

The term 'radiation' has two meanings. It can refer to the process of heat transfer, as well as the energy emitted in this process.

The radiation emitted by an object is an electromagnetic radiation, which can be classified into radio waves, microwaves, infra-red radiation, visible light, ultra-violet radiation, X-rays and gamma rays. You will learn more about this in Book 3B.

radiation 輻射

DSE goal

Realize the emission of infra-red radiation by hot objects.

a Emission of thermal radiation

All objects radiate energy. A hotter object radiates more energy. The type of radiation emitted by an object depends on its temperature. At room temperature, objects emit primarily **infra-red radiation** (Fig 2.2q). The hotter the object, the stronger the infra-red radiation it gives out.

If the temperature of an object further increases, the object may even emit visible light in addition to infra-red radiation (Fig 2.2r).

This image is taken and processed by a smartphone. Different colours in the image indicate that different amounts of infra-red radiation are emitted from various parts of the body, according to their temperatures.



Fig 2.2q An infra-red image of a man.

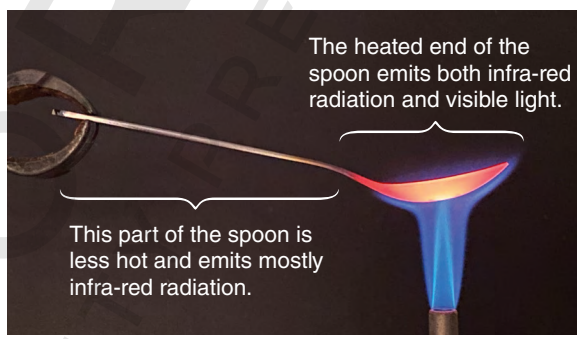
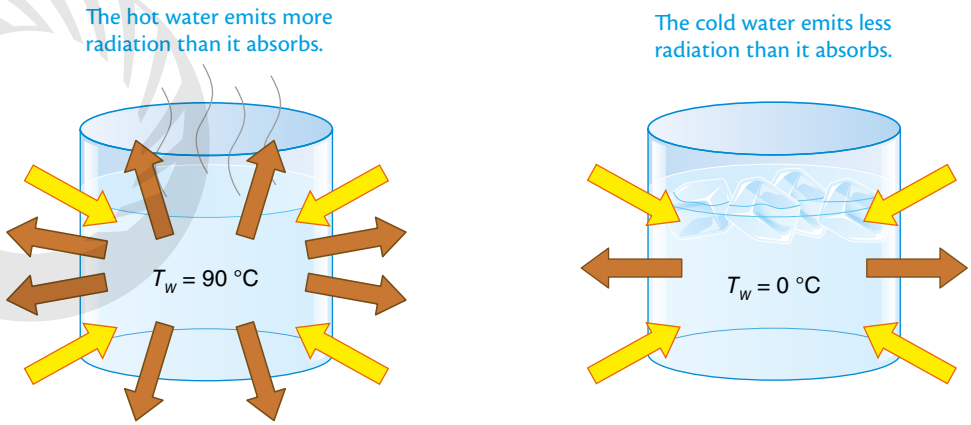


Fig 2.2r A metal spoon emits visible light when it is hot enough.

b Absorption and emission of radiation

An object not only emits radiation to the surroundings, it also absorbs radiation from the surroundings at the same time. Every object is both a **radiator** and an **absorber** of radiation.

In general, an object with higher temperature than the surroundings will have a net loss of energy by radiation and cool down. An object with lower temperature than the surroundings will have a net gain of energy by radiation and warm up (Fig 2.2s).



(i) A net loss of energy by hot water. *(ii) A net gain of energy by cold water.*

Fig 2.2s Net energy transfer by radiation between a glass of water and its surroundings at 25 °C (brown arrow: energy emitted by water; yellow arrow: energy absorbed by water).

infra-red radiation 紅外輻射　　radiator 輻射體　　absorber 吸收體

Google Forms Microsoft Forms

Quick check 7

→ E-version in OUP Exercise Platform is also available.

1 Which of the following heat transfer method(s) can take place in a vacuum? Put a tick in the corresponding box(es).

☐ Conduction ☐ Convection ☐ Radiation

2 *True or false:* Objects at 0 °C do not emit radiation. (T/F)

3 A flask filled with water at 75 °C is put in a room at 20 °C. Which of the following about the emission and absorption of radiation by the flask is correct?

A The flask only emits radiation to the surroundings and does not absorb radiation from the surroundings.

B The flask emits more radiation than it absorbs from the surroundings.

C The flask emits less radiation than it absorbs from the surroundings.

D The amount of radiation emitted and absorbed by the flask are equal.

DSE goal

Determine the factors affecting the emission and absorption of radiation.

DSE exam

13(1B)Q1(c), 22(1B)Q1, 23(1A)Q1

C Factors affecting the emission and absorption of radiation

There are seveal fators affecting how fast an object absorbs or emits radiation. Let us do the following experiments to see what they are.

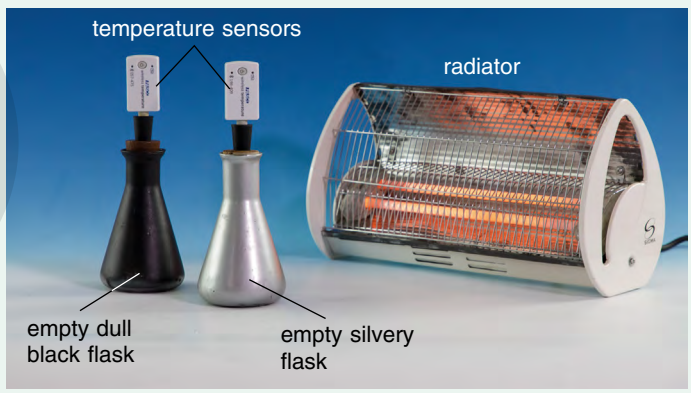
Video

→ This video demonstrates Expt 2a.

Experiment 2a Absorption of radiation by surfaces in different colours

1 Seal an empty dull black flask and an empty silvery flask using the stoppers fitted with temperature sensors. Record the temperature of air in the two flasks.

2 Place the two flasks at the same distance from a radiator. Turn on the radiator. After 10 minutes, record the temperature again.



Do not touch the heating part of the radiator unless it has cooled down.

→ This is a Video Quiz about another experiment investigating the absorption of radiation by surfaces in different colours.

Discussion

How does the surface colour of an object affect the absorption of radiation?

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Video

→ This video demonstrates Expt 2b.

Beware of the hot water.

Experiment 2b / Factors affecting emission of radiation

Part I Colour

1 Pour the same amount of boiling water into a dull black flask and a silvery flask.

2 Seal the flasks using the stoppers fitted with temperature sensors (Fig a).

3 Leave the flasks far apart on the table. After 10 minutes, record the temperature of water in the two flasks.



Fig a

Part II Temperature difference between an object and its surroundings

1 Pour the same amount of water into two dull black flasks, one at 50 °C and the other at 80 °C.

2 Seal the flasks using the stoppers fitted with temperature sensors (Fig b).

3 Leave the flasks far apart on the table. After 10 minutes, record the temperature of water in the two flasks.



Fig b

Discussion

How do the surface colour of an object and the temperature difference between the object and the surroundings affect the emission of radiation?

Simulation

→ This simulation shows the factors that affect the emission and absorption of radiation. It is available on OUP web in PC version (.exe) only.

i Colour

In Experiment 2a, the air temperature in the dull black flask rises faster. In Part I of Experiment 2b, the water temperature in the dull black flask drops faster. In fact,

- **dull black surfaces are good absorbers and good radiators of radiation;**
- **shiny and light-coloured surfaces are poor absorbers and poor radiators of radiation.**

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When radiation is directed onto shiny silvery surfaces, most of the radiation is reflected (Fig 2.2t). Therefore, they are good reflectors and poor absorbers of radiation. On the contrary, dull black surfaces are poor reflectors and good absorbers of radiation.

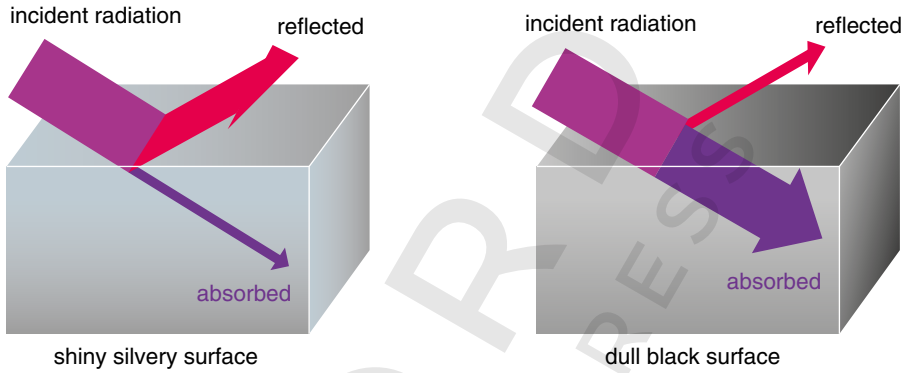


Fig 2.2t *Comparing the reflection and absorption of radiation between shiny silvery surfaces and dull black surfaces.*

The properties of dull black surfaces explain why we will feel hotter if we wear clothes in dark colours under strong sunlight (Fig 2.2u).

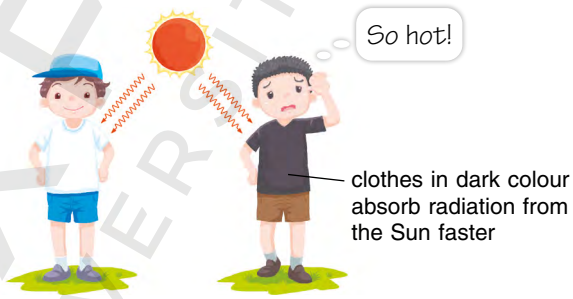


Fig 2.2u *Wearing clothes in light colours or dark colours?*

ii Temperature difference between object and surroundings

In Part II of Experiment 2b, we can see that the drop in temperature of the hotter water is faster. In fact,

> **the net rate of radiation emission or absorption increases when the temperature difference between the object and its surroundings increases.**

iii Surface area

The surface area of an object also affects its emission and absorption of radiation. In fact,

> **the rates of radiation emission and absorption both increase when the surface area of the object increases.**

Table 2.2a summarizes the factors that affect the emission and absorption of radiation.

Factor	Effect
Colour	dull black surface ⇒ good absorbers and good radiators
Temperature difference between the object and the surroundings	temperature difference ↑ ⇒ net rate of radiation emission or absorption ↑
Surface area	surface area ↑ ⇒ rate of radiation emission and absorption ↑

Table 2.2a *Factors affecting the emission and absorption of radiation.*

Google Forms Microsoft Forms

Quick check 8

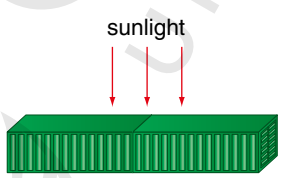
→ E-version in OUP Exercise Platform is also available.

1 A hot object with fixed mass is placed in a room. Which of the following can increase its net rate of radiation emission?
- Paint the surface of the object (dull black / shiny white).
- (Increase / decrease) the surface area of the object.
- Put it in a (warmer / cooler) room.

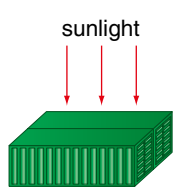
2 A cold object with fixed mass is placed in a room. Which of the following can increase its net rate of radiation absorption?
- Paint the surface of the object (dull black / shiny white).
- (Increase / decrease) the surface area of the object.
- Put it in a (warmer / cooler) room.

3 Two identical containers are placed within reach of sunlight. In which of the following situations would minimize the rate of increase in temperature inside the two containers?

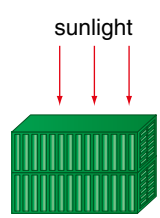
A



B



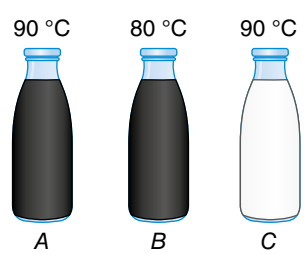
C



Learn by practice 6

1 'Wearing clothes in darker colours always makes you feel hotter.' Comment on this statement.

2 Three bottles contain the same amount of water at different temperatures. Bottles *A* and *B* are painted dull black while bottle *C* is painted white. They are all placed at room temperature. In which bottle does the temperature of water drop the fastest? Explain briefly.



d Examples of heat transfer by radiation

Designs that enhance or reduce heat transfer by radiation are commonly found in different applications in daily life (Fig 2.2 v). When the heat transfer by radiation is enhanced, the temperature of an object changes more quickly. On the other hand, when the heat transfer by radiation is reduced, the temperature change of the object can be minimized.

Enhancing radiation absorption



solar water heater
- It has numerous long tubes to provide large surface area
- It has tubes painted in black
  ⇒ water inside the tubes can be heated up quickly

Reducing radiation absorption



fuel storage tank
It is painted silvery white to avoid overheating by the radiation from the Sun

Enhancing radiation emission



car radiator
It is painted in black, so that it can lose energy and cool the engine quickly

Reducing radiation emission



foil blanket for marathon runner
It has shiny silvery surface to reduce energy loss by the runners

Fig 2.2v Examples of enhancing and reducing radiation.

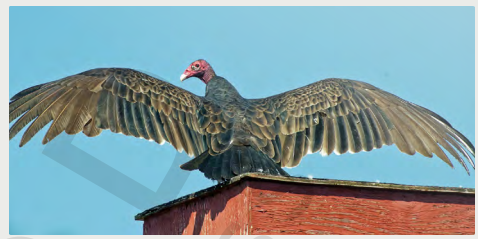
Know more

Heat transfer in nature

Animals may reduce heat gain or loss in different ways. Here are two examples.



A study has found that bird eggs laid in environment with intense sunlight tend to have shells in lighter colour. This avoid the eggs from being overheated by sunlight.



Turkey vulture is often seen standing in a spread-winged stance. It is believed that a vulture warms up its body in this stance by allowing sunlight shine on its feathers.

DSE goal

Identify the means of energy transfer in terms of conduction, convection and radiation.

DSE exam

14(1A)Q1, 15(1A)Q1, 16(1A)Q1, 19(1B)Q1(b), 20(1A)Q1

4 Examples with multiple ways of controlling heat transfer

In some applications, multiple designs are included to control the rate of heat transfer. The following are some examples.

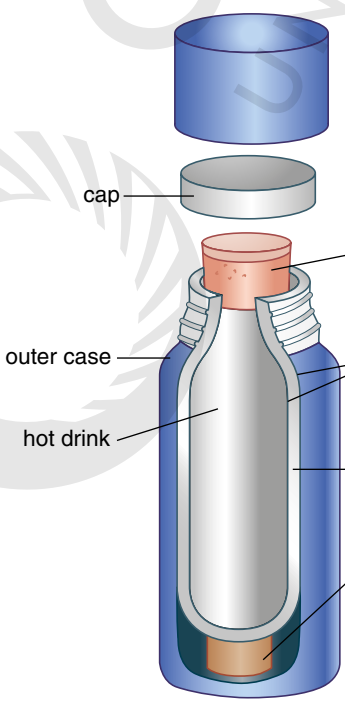
a Vacuum flasks

Video

→ This video shows the structure of a vacuum flask.

A vacuum flask (or thermos) keeps drink hot or cold by reducing the heat transfer between the drink and the environment through conduction, convection and radiation (Fig 2.2w).

✎ Ts may go through with Ss each feature of a vacuum flask.



Design	How it keeps drink hot
Plastic or cork stopper	reduce heat loss by conduction and convection
Silvery glass or stainless steel walls	reduce heat loss by radiation
Vacuum between double walls	prevent heat loss by conduction and convection
Insulated support	reduce heat loss by conduction

If cold drink is stored, the heat gain will be reduced.

Fig 2.2w *Vacuum flask and its designs that control heat transfer.*

Turkey vulture 土耳其禿鷹

☆ HKDSE 2016 Paper 1A Q1

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Thermal cup

A thermal cup filled with hot coffee has a silvery surface and a black lid, which seals the cup tightly. There is a vacuum layer between the inner and outer walls of the cup. Which of the following statements is/are correct?



(1) The lid reduces heat loss by convection and radiation.

(2) The silvery surface reduces heat loss by radiation.

(3) The vacuum between the walls reduces heat loss by conduction and radiation.

A (2) only B (3) only

C (1) and (2) only D (1), (2) and (3)

Solution

The lid can restrict the movement of the air heated by the hot coffee so heat loss by convection is reduced. However, the lid is black in colour and therefore is a good radiator of radiation.

∴ (1) is incorrect — Not all designs help in reducing heat loss.

Silvery surface is a poor radiator of radiation.

∴ (2) is correct

Radiation can take place in a vacuum and will not be slowed down by it. — **Common mistake** ⚠ Students may wrongly think that the vacuum can reduce heat loss by radiation.

∴ (3) is incorrect

The answer is A.

▶ Revision 2 Q7 (p.55)

b Greenhouse

A greenhouse is a special house that provides a warm environment for growing plants (Fig 2.2x). It is usually made of transparent materials such as glass.



Fig 2.2x Greenhouses.

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The following experiment shows how the change in air temperature is affected when air is trapped inside a glass container near a source of radiation.

Video

→ This video demonstrates Expt 2c.

Do not touch the hot incandescent lamp.

Experiment 2c / Heating and cooling of trapped air

1 Set up the apparatus as shown in Figure a. Make sure temperature sensors *A*, *B* and *C* are at the same distance from the incandescent lamp. Take the readings of the temperature sensors.

2 Switch on the lamp and take the readings again after about 10 minutes.

A filament lamp which emits primarily infra-red radiation should be used in this expt.

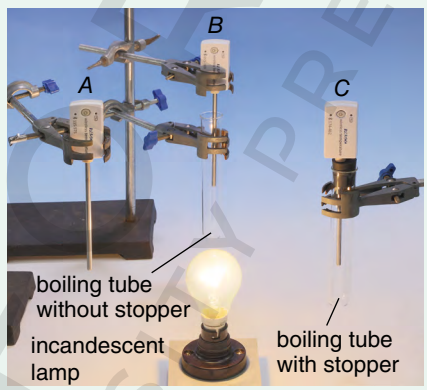


Fig a

3 Allow the temperature sensors to cool down so that their readings are about the same. Insert each temperature sensor into a boiling tube with a stopper and put them at equal distance from the lamp again.

Ts can also prepare the boiling tubes of air with about the same temperature by immersing them in a beaker of hot water for 5 to 10 minutes.

4 Switch on the lamp again. After about 10 minutes, record the readings of the temperature sensors and make sure that they are about the same.

5 Switch off the lamp. Remove the boiling tube and the stopper from temperature sensor *A* and remove the stopper from temperature sensor *B* (Fig b). Take the readings again after about 5 minutes.

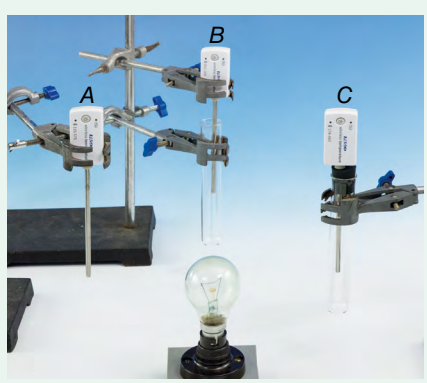


Fig b

Discussion

1 Which temperature sensor shows the biggest rise in temperature after the lamp is switched on?

2 Which temperature sensor shows the smallest drop in temperature after the lamp is switched off?

Experiment 2c shows that trapped air (inside the boiling tube with stopper) warms up more quickly and cools down more slowly than untrapped air. The effect is similar to that in a greenhouse.

The inside of a greenhouse is kept warm in the following way (Fig 2.2y).

① The radiation from the Sun enters the greenhouse.

② The contents inside the greenhouse absorb the radiation and become warm.

③ Warm objects in turn emit radiation, mostly in the form of infra-red radiation.

④ The infra-red radiation cannot pass through the glass and is trapped inside the greenhouse.

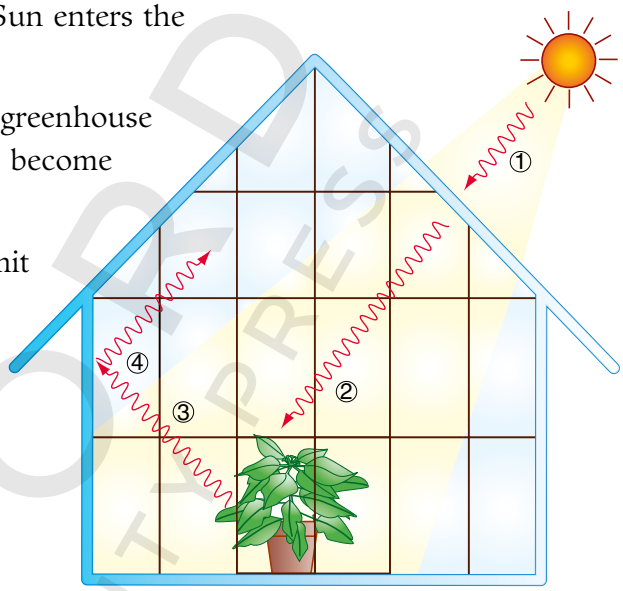


Fig 2.2y *Trapping energy inside a greenhouse.*

In addition, the warm air is (partly) trapped inside the greenhouse. This slows down the energy loss by convection. The combined effect keeps the entire greenhouse warm to protect plants in cold weather.

Know more

Igloos

In some extremely cold places like Greenland and Alaska, people use snow blocks to build *igloos*. How can igloos help keep people warm?

▸ p.58

STSE

The greenhouse effect and global warming

Do you know the relationship between the *greenhouse effect* and *global warming*?

▸ p.58

Know more

Heatstroke inside a car in the Sun

When a car is parked in the Sun, it becomes like a greenhouse (Fig a). As a result of the greenhouse effect, the temperature inside the car could reach 49 °C in a short time while the temperature outside is about 21 °C. With infra-red radiationing, a person in the car could suffer heatstroke.

Some people resolve this problem by propping up a silvery shade inside the windshield (Fig b). Can you explain how it works? Can you suggest other methods of reducing the greenhouse effect in a car?

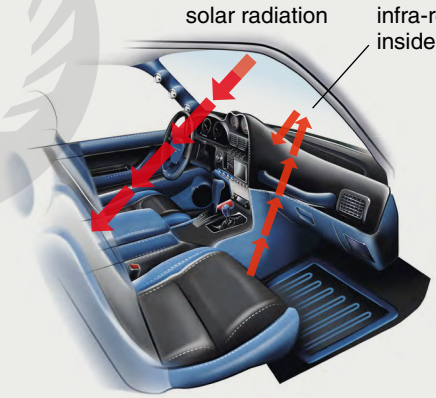


Fig a



Fig b

igloo 冰屋　greenhouse effect 温室效應　global warming 全球暖化

Quick check 9

→ E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

The following video demonstrates how a vacuum cooker works:

Video

1 The figure shows a vacuum cooker (Fig a). The inner pot with the food is first heated to a high temperature. It is then put inside the outer pot. After the lid is closed, the food inside the inner pot can be cooked continuously for a long time without heat source. Determine how the following designs in the vacuum cooker reduce energy loss. Put a tick in the corresponding box(es) in Table a.

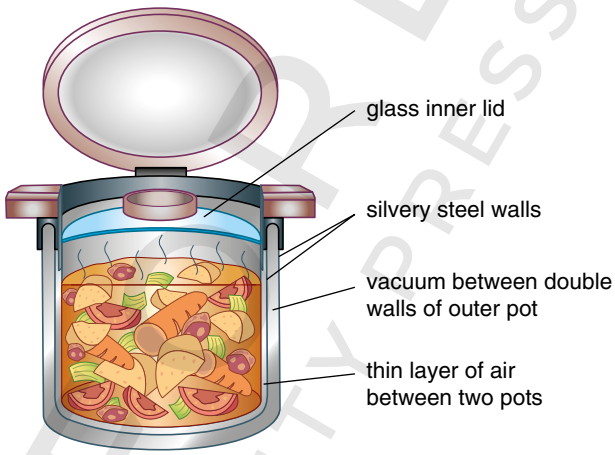


Fig a

Design	Type(s) of heat transfer reduced		
	conduction	convection	radiation
Glass inner lid			
Silvery steel walls			
Vacuum between double steel walls of outer pot			
Thin layer of air between two pots			

Table a

2 Figure b shows a greenhouse. On a sunny afternoon in winter, three thermometers *P*, *Q* and *R* were used to measure the air temperature at different positions shown in Figure b. Which one will record the highest temperature?

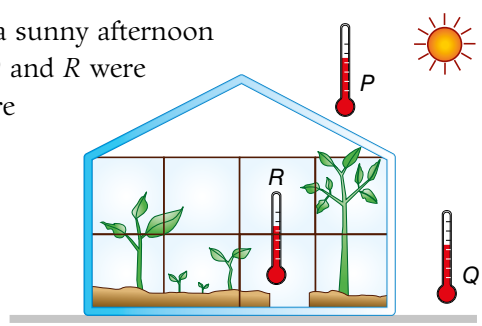


Fig b

3 How do the glass panels help in keeping the greenhouse warm?

A They do not allow most of the sunlight to enter the greenhouse.

B They do not allow most of the infra-red radiation to leave the greenhouse.

C They absorb most of the radiation and become warm.

Exercise 2.2

★**1** A spoon is put inside a cup of hot coffee (Fig a).



Fig a

Which of the following statements about the conduction involved are correct?

(1) Heat conduction takes place between the coffee and the spoon.

(2) Heat conduction takes place between the tip and the handle of the spoon.

(3) Heat conduction takes place between the spoon and the surrounding air.

A (1) and (2) only

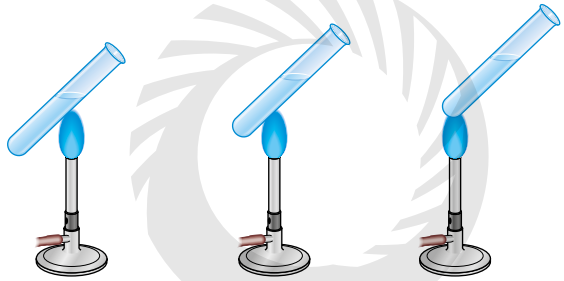
B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

★**2** A boiling tube containing water is heated with a Bunsen burner. Which of the following methods can heat up all the water in the shortest time?

(1) (2) (3)



A (1)

B (2)

C (3)

D They boil at the same time.

★**3** A radiation heater is turned on in a small sealed room built with dull grey concrete walls (Fig b). It is colder outside the room than it is inside.

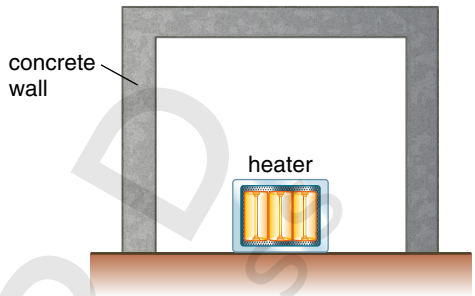


Fig b

Which of the following changes would help keep the room warmer?

(1) Paint the outer surface of the wall black.

(2) Paint the outer surface of the wall white.

(3) Replace the wall with a dull grey metal wall.

A (1) only

B (2) only

C (1) and (3) only

D (2) and (3) only

★**4** At a barbecue, some food is wrapped in aluminium foil and placed over the fire (Fig c).



Fig c

The food stays hot for a while even after it has been removed from the fire. The aluminium foil reduces energy loss from the food by

(1) conduction.

(2) convection.

(3) radiation.

A (1) only

B (3) only

C (2) and (3) only

D (1), (2) and (3)

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*5 A piece of T-shaped cardboard is put across the beaker (Fig d). A candle is put on one side of the cardboard and a burning incense stick is put on the other side.

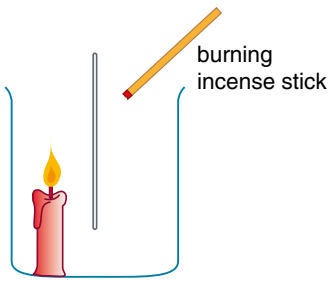


Fig d

When the candle is lighted up, the smoke from the incense stick is drawn by the convection current.

(a) Draw arrows to indicate how the smoke moves.

(b) Explain the formation of the convection current.

*6 Explain how the door seal (Fig e) in a refrigerator can help keep the food compartment cold.

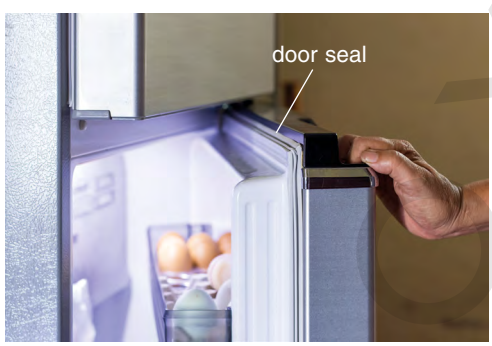


Fig e

*7 Figure f shows some freezers in a supermarket. Explain why they are effective in keeping the food cold even though they do not have covers.



Fig f

*8 Figure g shows a heat sink inside a computer.

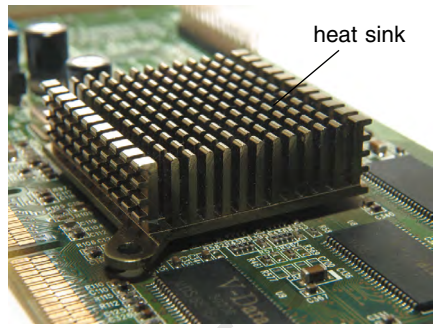


Fig g

Explain how each of the following designs of the heat sink helps dissipate heat.

(a) Made of metal

(b) Black in colour

(c) fin-like shape

*9 Figure h shows a lamp. Its lampshade has an opening at the top. What is the potential danger if the opening is covered? Explain briefly.

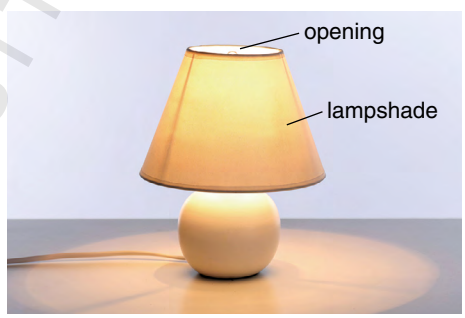


Fig h

*10 Figure i shows a cool box designed for keeping soft drinks cold in summer. Its **inner surfaces** are made of white foam.



Fig i

(a) Explain, in terms of the heat transfer by conduction, how the white foam keeps soft drinks cold.

(b) Explain, in terms of the heat transfer by radiation, how the white foam keeps soft drinks cold.

(c) Can this box keep hot drinks hot? Why?

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Key concepts 2

1 Molecular KE

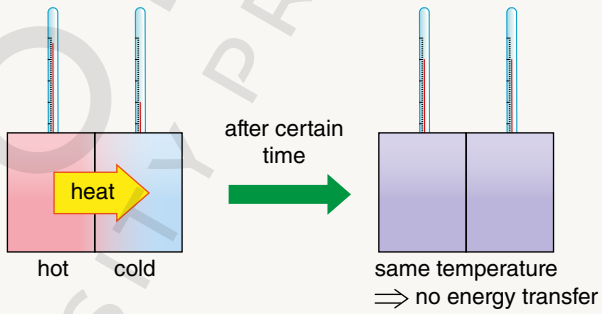
- Temperature is a measure of the **average** KE due to the random motion of the molecules in a body.
- Temperature ↑ ⇒ average KE ↑

2 Internal energy

- Internal energy is the total amount of the molecular energy (include molecular KE) stored in the body.
- Increase with temperature and mass

3 Heat

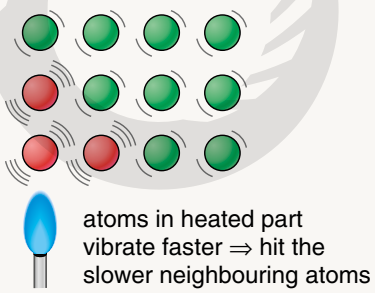
- Heat is the energy transferred from a hotter body to a colder body as a result of a temperature difference.



4 Heat transfer processes

Conduction

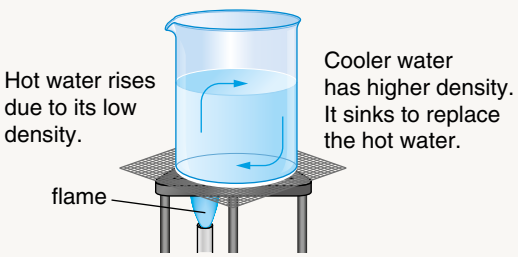
- Conduction is the transfer of heat from the hot part to the cold part of an object, or from a hot object to a cold object in physical contact.
- Energy (but not matter) is transferred.
- Good conductor: metals
  Good insulator: most non-metals
- Conduction in non-metallic solids



- Conduction in metals: collisions of atoms **plus collisions of free electrons**

Convection

- Convection is the process of heat transfer in a fluid by the actual movement of the fluid itself.



Radiation

- Radiation is a process of heat transfer which does not require any medium or particles.
- Radiation takes place in all directions.
- All objects emit and absorb radiation.
- Factors affecting emission and absorption of radiation: see Table 2.2a on p.44

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Key terms

1 absolute zero 絕對零度 p.22
2 absorber 吸收體 p.40
3 conduction 傳導 p.29
4 conductor 導體 p.30
5 convection 對流 p.36
6 convection current 對流 p.36
7 free electron 自由電子 p.34
8 heat 熱 p.25
9 infra-red radiation 紅外輻射 p.40
10 insulator 絕緣體 p.30
11 internal energy 內能 p.23
12 joule (J) 焦耳 p.23
13 kinetic energy 動能 p.20
14 radiation 輻射 p.39
15 radiator 輻射體 p.40
16 thermal equilibrium 熱平衡 p.26
17 work 功 p.26

Revision 2

Multiple-choice questions

★1 2.2 Some drinks are stored in a refrigerator at 4 °C (Fig a).



Fig a

When a canned drink and a drink in a plastic bottle are taken out of the refrigerator, which one feels colder? Why?

A They feel the same because their temperatures are the same.

B They feel the same because they have been placed in the same environment.

C The can feels colder than the plastic bottle because metal is a better conductor of heat.

D The plastic bottle feels colder than the can because plastic is a better insulator of heat.

★2 2.2 A pot of hot soup at 90 °C is placed in a closed space at room temperature. Which of the following statements is/are correct?

(1) Overall, the soup gains energy from the surroundings.

(2) The soup emits infra-red radiation.

(3) The soup stops emitting radiation when it reaches the room temperature.

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only

★★3 2.2 An aluminium tray contains some cold salad. The tray is covered with a white paper lid and left in a space at room temperature (Fig b).



Fig b

Which of the following about the heat transfer between the salad and the surroundings is/are correct?

(1) The aluminium tray reduces the heat transfer due to conduction.

(2) The aluminium tray reduces the heat transfer due to radiation.

(3) The paper lid reduces the heat transfer due to convection.

A (1) only B (2) only

C (2) and (3) only D (1), (2) and (3)

▶ Refer p.47

★★4 2.2 Three rods *X*, *Y* and *Z* have the same length. One end of each rod is immersed in a tray of iced water (Fig c). After a while, an infra-red image is taken (Fig d). The darker colours (violet and blue) indicate a lower emission of infra-red radiation.

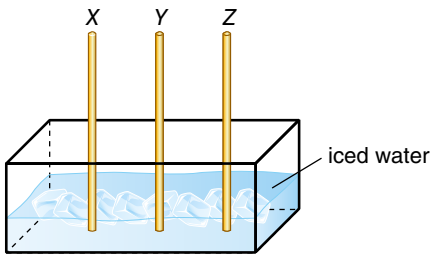


Fig c

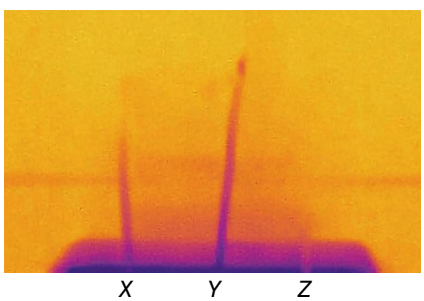


Fig d

It is known that one of the rods is a glass rod, one is a copper rod and one is an iron rod. Which of the following correctly matches the rods with the materials?

	Rod *X*	Rod *Y*	Rod *Z*
A	copper	iron	glass
B	iron	copper	glass
C	iron	glass	copper
D	glass	copper	iron

Refer p.30, 43

2.2 **5** **HKDSE 2014 Paper 1A Q1**



Fig e

Two identical scoops of ice cream are transferred from a refrigerator into paper cup *X* and vacuum flask *Y* shown above. Under room temperature, the time required for the ice cream in the containers to melt completely is t_X and t_Y respectively. What is the expected result and explanation?

A $t_X > t_Y$ as the vacuum flask reduces heat loss to the surroundings.

B $t_X > t_Y$ as the vacuum flask retains the heat.

C $t_Y > t_X$ as the vacuum flask keeps things cold by releasing heat into the surroundings.

D $t_Y > t_X$ as the vacuum flask reduces the rate of heat gain from the surroundings. (83%)

2.2 **6** **HKDSE 2015 Paper 1A Q1**

A driver turns off the engine after parking his car outdoors under the Sun. Two hours later when getting into the car, he feels that the inside of the car is far hotter than outside. The best explanation is

A the car's engine is still generating heat after the engine has been switched off.

B the car's metal parts absorb infra-red radiation at a faster rate than the surroundings.

C the glass windows of the car trap infra-red radiation and a greenhouse effect results. (66%)

D the surrounding air is a good insulator of heat which reduces heat loss by conduction.

2.2 **7** **HKDSE 2016 Paper 1A Q1**

Some icy cold liquid is kept cold inside a vacuum flask. Which statements are correct?

(1) The flask's cork stopper reduces heat gain from the surroundings.

(2) The silver coating on the inner surface of the glass wall is a good reflector of infra-red.

(3) The vacuum between the double glass walls reduces heat gain by radiation.

A (1) and (2) only (59%)

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

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2.2 **8** **HKDSE 2020 Paper 1A Q1**

A heater is installed at the centre of a fully filled cubic water tank. Temperature sensors *P*, *Q*, *R* and *S* are fixed at the respective centres of the top, left, right and bottom surfaces of the tank.

After the heater is switched on for **a short duration**, which pair of sensors below would indicate the largest temperature difference?

A *Q* and *R*

B *R* and *S*

C *Q* and *S*

D *P* and *R* (56%)

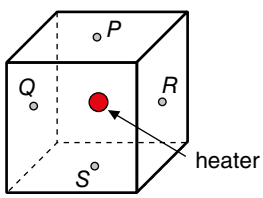


Fig f

Conventional questions

*9 2.2 Jerry places some ice cream scoops in a metal cup and a paper cup (Fig g and h).



Fig g



Fig h

(a) Explain, in terms of molecular motion, how energy is transferred by conduction between the paper cup and the ice cream scoops. (3 marks)

(b) Explain why the ice cream scoops melt more quickly in the metal cup than in the paper cup. (2 marks)

*10 2.2 Some ice is kept at the bottom of a boiling tube by a piece of wire gauze under water. The upper part of the boiling tube is heated with a Bunsen burner. The water at the top is boiling but the ice trapped at the bottom melts very slowly (Fig i).

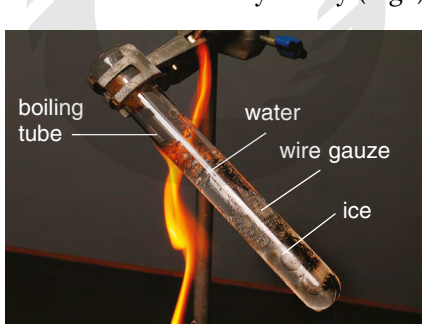


Fig i

(a) Explain briefly why the ice does not melt quickly. (2 marks)

(b) Explain how the time needed for the ice to melt would be affected if the following changes are made separately.

(i) The glass boiling tube is replaced with a metal tube.

(ii) The lower part of the boiling tube is heated instead of the upper part. (6 marks)

**11 2.2 Paul uses the following set-up (Fig j) to verify that a black surface is a good radiator. *A* and *B* are two identical beakers except that *B* is painted black. Initially, they contain the same amount of hot water at the same temperature.

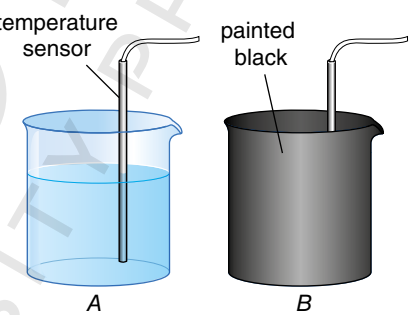


Fig j

Figure k shows the temperature variation in beaker *A*.

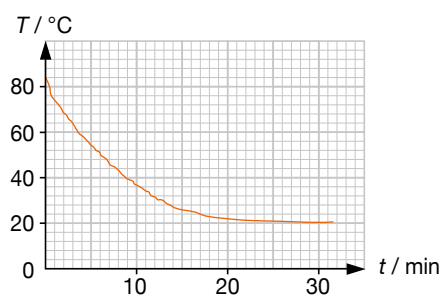


Fig k

(a) What is the temperature of the space in which the set-up is placed? (1 mark)

(b) Why does the water temperature in beaker *A* drop faster at the beginning than later on? (2 marks)

(c) In Figure k, sketch the expected temperature variation of the water in beaker *B*. (2 marks)

(d) How does the curve in (c) verify that a black surface is a good radiator? (2 marks)

(e) How would you modify Paul's set-up to verify that a black surface is a good radiation absorber. (2 marks)

Refer p.43–44

12 ★★ 2.2 Figure l shows a model which demonstrates how a solar water heater works. When the model is put under sunlight, the water in the tube will be heated up and move back to the plastic bottle.

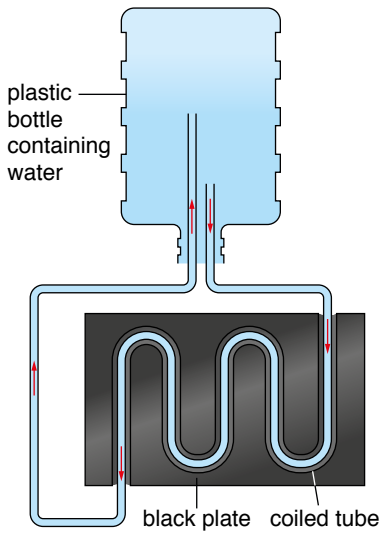


Fig l

(a) Suggest an advantage of using a plate black in colour. Explain your answer. (2 marks)

(b) Suggest and explain whether the plate should be made of plastic or copper. (3 marks)

(c) Suggest an advantage of using a coiled tube in this model. Explain your answer. (2 marks)

(d) A student suggests that the water can be heated up more quickly if the plate is sealed in a transparent glass box. Comment on this suggestion. (3 marks)

▶ Refer p.49

13 2.2 **HKDSE 2019 Paper 1B Q1(b)**

Some ice cream at –10 °C is put into a 'thermal bag', of which the inner layer is made of polyethylene foam coated with aluminium foil. The bag is also equipped with a zipper at the top (Fig m).



Fig m

The thermal bag is then brought outdoors on a hot sunny day.

(a) Referring to the heat transfer processes, explain **one** feature of this bag that helps keep the ice cream at a low temperature. (1 mark)

(b) Suggest **one** modification to this bag that would enhance its ability to keep things stored inside at a low temperature. (1 mark)

14 ★★ 2.2 Read the following passage about **a mug** and answer the questions that follow.

A mug is a very common kind of cup. It is usually made of ceramic. The bottom surface of a mug is usually concave in shape (Fig n). This effectively reduces energy transfer between the mug and the table by conduction. The mug handle is also made of ceramic and has small contact areas with the body of the mug. Hence, the handle heats up unnoticeably even if the drink in the mug is very hot. With these features, most of the heat transfer is through the surface of the drink. To further reduce heat loss, you can choose a mug with a smaller opening or painted in a light colour. Of course, you can also cover your mug with a lid.

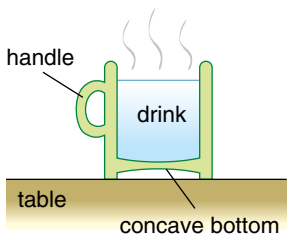


Fig n

(a) How does a lid help reduce the heat transfer between the drink and the surroundings if the drink is (i) hot, (ii) cold? (3 marks)

(b) Suggest a colour for the lid so that it can best maintain the temperature of a drink. (1 mark)

(c) Explain why a small opening helps reduce energy loss. (1 mark)

(d) Consider a ceramic mug and a metal mug of the same size and shape. Which better maintains the temperature of a drink? Why? (2 marks)

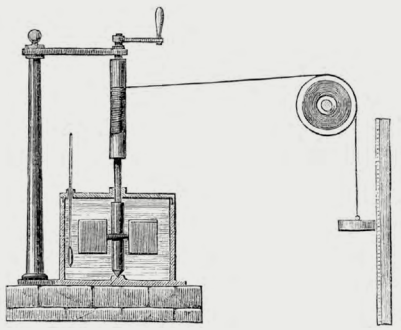
▶ Refer p.45–46

Q13 Exam Report: The heat transfer processes were well understood by candidates, however, quite a number of them mistook the direction of heat flow and stated that the inner silvery surface was a poor absorber of radiation in (b).

Know more

James Joule (1818–1889) (p.25)

James Joule was an English physicist. By using apparatus similar to that shown in the figure, Joule showed that 'heat' was a form of energy and not a substance. His findings were further developed into the law of conservation of energy, which became one of the most important principles in physics. The unit of energy is named after him.



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Know more

Igloos (p.49)

In some extremely cold places like Greenland and Alaska, temperatures can be as low as −45 °C. Some people there use snow blocks to build igloos (Fig a). Figure b shows some features of an igloo which help to keep the temperature inside the igloo higher than the temperature outside.



Fig a

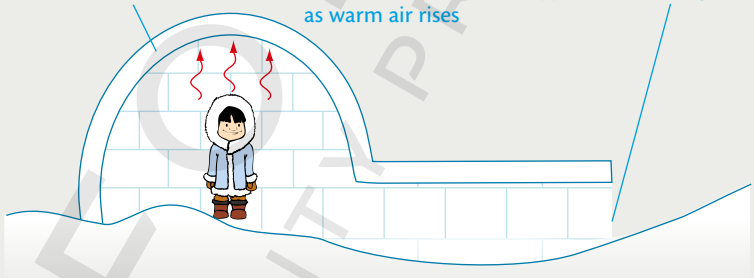


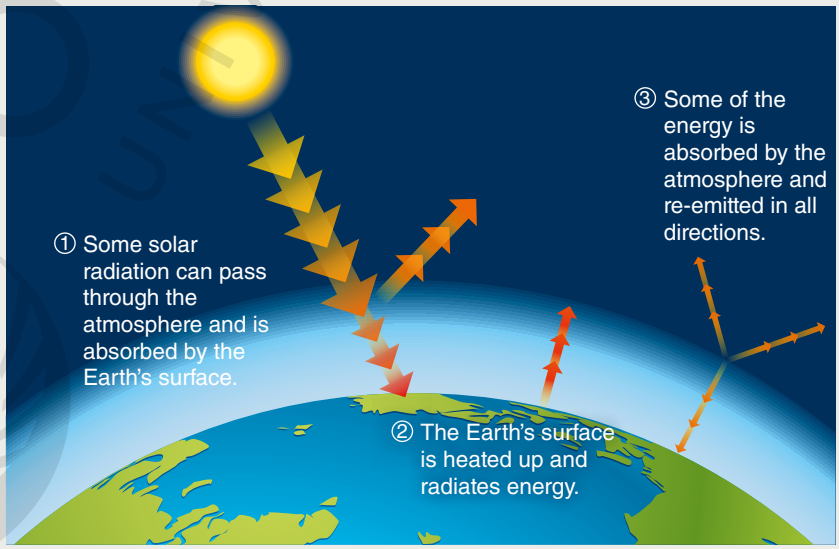
Fig b

OUP eResources

STSE

The greenhouse effect and global warming (p.49)

The natural greenhouse effect keeps the Earth warm and makes it a suitable place for life. The following diagram explains how the natural greenhouse effect occurs. What would happen to the Earth without the natural greenhouse effect?



Since the 19th century, more and more greenhouse gases, such as carbon dioxide, have been emitted. These greenhouse gases mainly come from human activities, particularly burning fossil fuels. Excessive amounts of greenhouse gases enhance the greenhouse effect, causing the average surface temperature of the Earth to increase. This climate change is called global warming.

It is believed that global warming may lead to more extreme weather, such as severely high or low temperatures and heavy rainfalls. Extreme weather events in Hong Kong have generally become more frequent. What are the potential impacts of such a change on our lives?

To learn more about extreme weather events in Hong Kong, visit the Hong Kong Observatory website.

Useful link